

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.2: Chemical Families — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.2: Chemical Families
Driving Question	DRIVING QUESTION: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm — and how do their "family" properties explain the materials that build, light up, and sustain our daily lives?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Family Resemblances: Launching the Chemical Families Phenomenon	Students observed four anchoring phenomenon objects: a sodium street lamp photograph from Thika Road, a jua kali argon-welding image from Kamukunji, a calcium hypochlorite tablet fizzing in water, and a helium party balloon. Students observed that these four substances look completely different, behave completely differently, and are used for completely different purposes — yet all are elements.	Students learned the term chemical family as a label for a group of elements with similar properties due to similar electron arrangements in their outer shells. Students recalled from Sub-Strand 2.1 that elements have electrons arranged in shells and that the outer shell is significant. Students learned that the class has a Driving Question Board that they will update over eight lessons as they gather evidence.	Students cannot yet fully explain why the four phenomenon elements behave differently — this is intentional. Students have articulated their first-draft model: an initial grouping of 20 elements based on perceived similarity, and an initial characterisation of the four phenomenon elements. The gap between what students observed and what they can explain is the productive tension that drives the remaining seven lessons.
Lesson 2 Sorting the Periodic Table Into Families: Who Belongs Together?	Learners sorted 20 element cards into family groups based on electron arrangement patterns and observed that elements sharing the same number of valence electrons could be placed in the same column. They observed that sodium (street lamp), potassium (Rift Valley fertiliser), and lithium all have 1 valence electron; that helium and argon (welding shield, party balloon) both have full outer shells; and that chlorine has 7 valence electrons — one short of a full shell.	Learners learned that the periodic table is organised into chemical families — alkali metals (Group 1), alkaline earth metals (Group 2), halogens (Group 7), noble gases (Group 0), and transition metals — and that membership in a family is determined by the number of valence electrons. Elements in the same family have the same number of valence electrons and therefore behave similarly. The dramatically different behaviours of sodium (reactive), argon (inert), chlorine (reactive, disinfecting), and helium (inert) seen in the anchoring phenomenon can be explained by their	The anchoring phenomenon (sodium lamp glowing orange, argon shielding a Kamukunji weld, chlorine compound fizzing in a water tank, helium floating inertly in a balloon) is explained by valence electron count: sodium's single valence electron makes it highly reactive and able to release energy as light in a discharge lamp; argon's and helium's full outer shells make them completely unreactive — perfect as protective or inert gases; and chlorine's 7 valence electrons make it 'eager' to gain one more, driving its strong reactivity with water and its disinfecting power. Same

		different family memberships.	family = same valence electrons = similar behaviour.
Lesson 3 Alkaline Earth Metals — Moderate Reactivity and the Chemistry Behind Kenyan Fertilisers	Magnesium ribbon reacted with dilute hydrochloric acid producing vigorous bubbling — hydrogen gas confirmed by squeaky pop with a lit splint — and the ribbon dissolved to form a colourless solution of magnesium chloride. Magnesium also reacted with steam (teacher demonstration) producing a white powder (magnesium oxide) and hydrogen gas, but the reaction with steam was slower than with acid. Magnesium did not react noticeably with cold water during the observation period.	Alkaline earth metals (Group 2) have two valence electrons and form 2+ ions when they react. Having two electrons to remove requires more energy than removing one, so Group 2 elements are less reactive than Group 1. Reactivity increases down Group 2. Calcium and magnesium are used in CAN fertiliser for Kenyan farming because their controlled reactivity is agriculturally useful without being hazardous.	The difference in reactivity between sodium (Group 1, one outer electron) and magnesium (Group 2, two outer electrons) is explained by the energy needed to remove outer electrons: removing two electrons is harder than removing one. This electron-arrangement pattern also explains why Group 2 elements can be safely used in fertiliser bags stored in agrovet shops and spread on Rift Valley maize fields — their moderate reactivity makes them useful, not dangerous. This advances the driving question: the number of valence electrons determines how reactive an element is, which determines what it can safely and usefully do in daily Kenyan life.
Lesson 4 Alkali Metals in Action: Why Does Reactivity Increase Down Group 1?	Students watched a teacher demonstration (or verified video) of lithium, sodium, and potassium reacting with water and recorded observations on a structured Group 1 Observation Grid — noting increasing speed of reaction, fizzing (hydrogen gas), movement on the water surface, and the production of an alkaline solution confirmed by phenolphthalein turning pink. The observation that potassium produced a lilac/violet flame while sodium did not was noted as a key qualitative difference.	Students learned that all Group 1 alkali metals have exactly 1 outer-shell electron; that this outer electron is lost more easily as atomic radius increases going down the group (because the outer electron is further from the nucleus and shielded by more inner electron shells, reducing the effective nuclear attraction); that this explains why reactivity increases $\text{Li} < \text{Na} < \text{K}$; and that the reaction of any alkali metal with water follows the pattern: metal + water → metal hydroxide + hydrogen gas, with balanced equations $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$ and $2\text{K} + 2\text{H}_2\text{O} \rightarrow 2\text{KOH} + \text{H}_2$. Students also connected sodium's loosely held outer electron to its behaviour in street lamp tubes — the same electron property drives both chemical reactivity and the characteristic orange-yellow light emission.	Students constructed a class consensus causal explanation: 'All Group 1 metals have 1 outer-shell electron. Going down the group, each successive element has one more electron shell, placing the outer electron further from the nucleus. The electrostatic attraction between the nucleus and the outer electron therefore decreases, making the electron more easily lost. A more easily lost outer electron means the metal reacts more vigorously with water, producing metal hydroxide and hydrogen gas faster. This is why potassium reacts most vigorously of the three, sodium intermediate, and lithium least vigorously — and it is why sodium inside a Thika Road street lamp tube, though not reacting chemically, readily absorbs electrical energy through that same loosely held outer electron and re-emits it as distinctive orange-yellow light.'
Lesson 5 Noble Gases: Why Do Some Elements Do Absolutely Nothing?	Students examined two projected images side by side: a jua kali welder on River Road, Nairobi, surrounded by a visible shimmer of argon gas shielding his torch, and a Kenya Meteorological Department technician in Dagoretti releasing a helium	Students established that noble gases — helium, neon, argon, krypton, and xenon — belong to Group 18 and each possesses a completely full outer electron shell, which means they have no tendency to gain, lose, or share electrons with other atoms. This	The anchoring phenomenon asks why the argon-shielded jua kali welding flame burns with intense heat yet leaves no ash and does not react with the metal being welded, and why a helium party balloon is completely harmless. This lesson provides

	weather balloon. During the guided research phase, groups recorded data from textbook pages, the periodic table poster, and approved websites into a structured table covering the name, symbol, outer-shell electron count, a physical property, and one verified Kenyan use for each of the five noble gases — helium, neon, argon, krypton, and xenon. Students also observed the teacher draw electron shell diagrams for argon (2, 8, 8) alongside the already-familiar diagrams for potassium (2, 8, 1) and chlorine (2, 8, 7) on the board. Finally, each student completed a Model Card, first filling in all five noble gases from memory and then checking their entries against their research cards, marking corrections in a different colour pen.	full-shell configuration is the direct cause of their chemical inertness: because their outer shells are already stable, they do not form bonds or undergo the reactions observed in alkali metals and halogens in earlier lessons in this unit. Students recorded the key vocabulary of chemical inertness and linked it to the phrase full outer shell. They also learned that inertness does not mean uselessness: neon is used in road signage along Mombasa Road, argon shields metal in jua kali welding, helium lifts weather balloons for the Kenya Meteorological Department, and krypton is used in photographic flash lamps, with each application relying specifically on the gas being chemically unreactive.	the electron-level explanation: argon and helium have full outer shells, so unlike sodium — which loses its single outer electron to produce the orange glow of the Thika Road street lamps — these gases have no electron to donate, accept, or share, making chemical reaction impossible under ordinary conditions. The comparison of the argon shell diagram (2, 8, 8) with the potassium diagram (2, 8, 1) studied earlier in this unit made the contrast concrete: one extra electron on the outer shell of potassium drives dramatic reactivity, while a completely filled shell in argon drives complete stability. Students updated the Driving Question Board to reflect that full outer shells explain both inertness and specific industrial utility, advancing the unit-level explanation of why elements from different chemical families behave so differently in the materials that build, light up, and sustain Kenyan daily life.
Lesson 6 Magnesium in Action: Group 2 Reactivity, Fertilisers, and the Two-Electron Difference	Magnesium ribbon placed in cold water produced very slow, tiny bubbles and a faint white film ($\text{Mg}(\text{OH})_2$) — the universal indicator barely changed. Magnesium in dilute HCl produced rapid, vigorous bubbling; the gas tested positive for hydrogen (squeaky pop). Magnesium ribbon held over steam reacted moderately, producing a white solid (MgO) and hydrogen gas. In all three cases, the reaction was noticeably slower and less dramatic than the sodium-in-water demonstration from Lesson 3.	Magnesium (Group 2, electron configuration 2,8,2) has two outer electrons. Both must be lost to form the Mg^{2+} ion. Because the nucleus attracts two outer electrons more effectively than sodium's nucleus attracts its single outer electron, it is harder for magnesium to lose electrons — making Group 2 elements less reactive than Group 1 elements. Balanced equations: $\text{Mg}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$; $\text{Mg}(\text{s}) + \text{H}_2\text{O}(\text{steam}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2(\text{g})$. Calcium (also Group 2) is the active ingredient in CAN fertiliser (Calcium Ammonium Nitrate) used by Rift Valley maize and wheat farmers — its controlled reactivity means it releases Ca^{2+} ions steadily into the soil rather than reacting explosively.	The anchoring phenomenon asks why some elements react violently and others do not. Today's evidence adds a new layer: it is not just WHICH group an element is in, but HOW MANY outer electrons it has and how tightly the nucleus holds them. Sodium (1 outer electron, Group 1) loses it easily → violent reaction with water → orange glow in a Thika Road street lamp. Magnesium (2 outer electrons, Group 2) loses them more reluctantly → slower, controlled reactivity → safe for agricultural use in CAN fertiliser on Kenyan farms. The same periodic table position that makes sodium spectacular makes calcium useful — chemical family properties are engineered into the materials that sustain Kenyan daily life.
Lesson 7 Transition Metals: The Workers of the Periodic Table — Iron, Copper, and Zinc in Kenyan Life	Learners observed the physical properties (lustre, malleability, electrical conductivity, high melting points) and chemical reactivity (slow reaction of iron and zinc with dilute HCl; no reaction of copper with dilute HCl; copper turning black on heating in air) of iron, copper, and zinc samples. They also	Transition metals form a distinct chemical family with high melting points, hardness, malleability, and good electrical and thermal conductivity. Unlike alkali metals (sodium), they react slowly or not at all with cold water, and react slowly with dilute acids. They form coloured compounds (e.g., blue	The contrasting behaviours of sodium (Group 1 alkali metal — explosive reactivity) and iron/copper/zinc (transition metals — slow, controlled reactivity) are explained by their positions in the periodic table and the resulting differences in electron configuration and ionisation energy.

	observed video evidence of copper oxidation in a flame, contrasting with the inert argon used in the welding torch from the anchoring phenomenon.	copper(II) sulphate, green iron(II) chloride) and have variable oxidation states. These properties directly explain why iron is used for jembes, copper for electrical wiring, and zinc for coating mabati roofing sheets in Kenya.	Transition metals, sitting in the d-block, have higher nuclear charge and more complex electron structures that make them less reactive than Group 1 metals but still reactive enough to be chemically useful. This explains why the jembe (iron) rusts slowly over months rather than exploding instantly like sodium, and why copper wiring conducts electricity reliably without reacting with air under normal conditions — linking directly to the anchoring phenomenon of the Thika Road street lamp (sodium), the welding torch (argon), and extending the explanation to include the 'silent worker' metals of Kenyan daily life.
Lesson 8 Final Explanation: Why Do Chemical Families Determine How Elements Build, Light Up, and Sustain Our Daily Lives?	Students observed (across Lessons 1–7, reviewed in the Gallery Walk): sodium fizzing and burning violet in water; the orange-yellow glow of a sodium vapour discharge tube (modelling Thika Road street lamps); a hibiscus flower bleaching white in chlorine water (modelling Nairobi Water chlorination); iodine solution turning blue-black on ugali flour paste (connecting to Strand 1.3 food tests); the inertness of noble gases in sealed discharge tubes (modelling the argon welding shield and helium party balloon); and the slow rusting of iron (modelling the jembe/hoe and mabati corrugated iron roof on a Kenyan farm).	Students learned that elements belong to chemical families based on their electron configuration and position in the periodic table. Alkali metals (Group 1, e.g. sodium) have one valence electron and react vigorously with water, producing metal hydroxides and hydrogen gas. Alkaline earth metals (Group 2, e.g. magnesium, calcium) have two valence electrons and are less reactive than Group 1 but still reactive with oxygen and acids. Halogens (Group 17, e.g. chlorine, iodine) have seven valence electrons, are highly electronegative, and react with water to form bleaching/disinfecting solutions; they show a colour trend down the group (F_2 pale yellow gas $\rightarrow$ Cl_2 yellow-green gas $\rightarrow$ Br_2 red-brown liquid $\rightarrow$ I_2 grey solid). Noble gases (Group 18, e.g. helium, argon, neon) have complete outer shells, are chemically inert, and produce characteristic coloured glows when electricity passes through them in sealed tubes — with no chemical change occurring. Transition metals (e.g. iron, copper, zinc) have variable oxidation states, are hard and dense, conduct electricity, form coloured compounds, and corrode slowly (unlike alkali metals which react explosively). Every observed behaviour traces back to the number and arrangement of valence electrons.	Students explained that the anchoring phenomenon — sodium street lamps glowing orange on Thika Road, argon shielding a jua kali weld, chlorine compounds purifying Nairobi Water, and helium floating inertly in a party balloon — is fully accounted for by chemical family membership and electron configuration. Sodium glows because it is a Group 1 alkali metal: its single valence electron is excited by electrical energy in the discharge lamp and emits orange-yellow light when it returns to ground state; on Thika Road, thousands of sodium atoms do this simultaneously every night. Argon shields the weld because it is a Group 18 noble gas with a full outer shell (eight electrons), making it chemically inert — it neither reacts with the hot metal nor with atmospheric oxygen, providing a protective blanket with no chemical consequences. Chlorine (Group 17, seven valence electrons) reacts with water via $Cl_2 + H_2O \rightarrow HCl + HClO$; the hypochlorous acid (HClO) produced is a powerful oxidising bleach that destroys the chromophores in the hibiscus petals and kills pathogens in Nairobi Water storage tanks. Helium (Group 18, full outer shell with two electrons) is completely inert and non-toxic, making it the safe choice for party balloons. Iron (a transition metal) corrodes slowly via $4Fe +$

			$3\text{O}_2 + 6\text{H}_2\text{O} \rightarrow 4\text{Fe}(\text{OH})_3$ because its partially filled d-orbitals give it moderate reactivity — far less than sodium, but more than noble metals. The periodic table is therefore a map of chemical behaviour: knowing an element's family tells you how it will behave before you ever touch it.
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